

However, increased automation and intelligence could potentially compromise cybersecurity. The increasingly large volumes of data that is necessary for Artificial Intelligence bring along with its privacy and security concerns. With the rise of Quantum Computing, it will become necessary to change the way the data is encrypted, and communications will be combined with quantum-resistant technologies to enhance cybersecurity.

An exponential growth in IoT (Internet of Things) devices will help create a hyperconnected world. It will be a trillion-dollar industry with billions of connected devices which will enable increased efficiency, automation, convenience, and personalization. Increased connectivity will change the way individualized products are developed, thereby influencing the production process of large enterprises in particular.

Industry 4.0 will restructure the manufacturing process and eliminate unwanted human interference. Industry 5.0, following shortly, would enable collaboration between industrial automation and human thinking skills. An increasing consumer-demand for personalization will bring about extensive hands-on customization of the products. Industry 5.0 will create a synergistic relationship between humans and machines, enabling companies to take processes to levels of seamlessness and efficiencies previously unheard of.

Intervention of technology in Customer Experience will realign the way products and services are consumed. Consumers will be subject to a hyper-personalized experience augmented with Artificial Intelligence insights leading to an enhanced customer-journey.

Augmented Reality, incorporating digital sensory stimuli with the real world, will alter the consumer experience, allow 3D visualization of organs and aid doctors, make education more interactive, allow consumers to imagine the products and beyond.

Consumers of tomorrow will demand hyper-personalization with the increasing use of AI, AR/VR

(Augment Reality/ Virtual Reality) and expect an autonomous experience. Many global brands have started testing the autonomous functionality of their products and quite soon, experiences such as autonomous shopping or autonomous driving will become the order of the day.

## Technology industry in the next 20 years

- **Social media:** Experts observe a reckoning for social platform companies, which will enable large-scale changes in the overall environment of social media during the next decade or so. Either there will be a reckoning for technology companies producing major revisions to their platforms or there could be new platforms designed to make their users' best interests paramount. It is expected that new social platforms will focus on privacy and 'genuine' information.
- **Privacy issues & misinformation:** Actions will be taken to ring-fence online privacy. Online advertising regulations will become stringent; data privacy and protection will assume high importance. All sections of the society will have access to affordable tools and technologies to protect their privacy. There are growing concerns about the accuracy of information encountered online, hence actions are necessary to curb online visibility of false or misleading information. Some amount of change can be brought about by ushering in digital and technology literacy.
- **Healthier living:** Innovations will address physical and mental health; enabling major changes for the healthcare industry and professionals. Experts predict significant medical advancements in the next decade and expect innovations in every realm of physical and mental health. These changes are likely to improve the standards of health for common individuals.
- **Artificial intelligence:** AI will continue to grow and have an increasing impact on human lives. Algorithms have been advancing

over the years. New organizational technologies that are being catalyzed viz. 'software-defined organizations' will focus on combining the resources available on digital platforms to create and deliver continuous business-value.

- **Educational reform:** Education systems will evolve and transform in response to the multilayered societal changes that are likely to prevail. As society shifts under the influence of technology, experts foresee education changing in response. A greater focus on science, technology, engineering and math (STEM) subjects is expected coupled with Digital literacy programs in the standard curriculum. It seems quite likely that the Education systems would transform and become increasingly digital-oriented.
- **Labor and jobs:** Business practices, work lives and the larger economy will substantially change in the next decade or two. In the light of growing technological developments, it is expected that innovations are likely to occur in labor management influencing the jobs market. From shorter work-weeks to technological displacement, a number of changes are expected over the next 20 years.
- **Environmental issues:** Experts suggest that climate change and other environmental issues will inspire innovation out of necessity. It is quite likely that Digital technologies and Digital methodologies are likely to help effect this change.

## Other Technologies that are likely to bring about a profound impact in the upcoming years include:

- Blockchain
- Cryptocurrency
- AI Cloud Services
- Smart Homes
- 5G
- Autonomous Vehicles

It is important to note that technology by its nature is value-neutral; and it is what we do with it that precisely defines whether we are creating the "next big thing of value" that can enable wider benefits. 